

S $\Delta\phi$ -08

Agency, Responsibility, and Freedom (ARF)

Conditional Closure of Recursive Transition Systems

Version v1.0

Sofience- $\Delta\phi$ Formalism Series

This document states a conditional binding over prior minimal axioms in the series. It does not assert necessity by definition; instead it specifies minimal assumptions under which the ARF triangle is the smallest stable description of recursive transition systems.

Series references (DOI):

- <https://doi.org/10.5281/zenodo.18731366> — S $\Delta\phi$ -00 — The Sofience- $\Delta\phi$ Formalism: Conceptual Architecture and Series Overview (v1.0)
- <https://doi.org/10.5281/zenodo.18730265> — S $\Delta\phi$ -01 — Sofience- $\Delta\phi$ Minimal Core: Phase-Change Formalism of Existence (v1.1)
- <https://doi.org/10.5281/zenodo.18730296> — S $\Delta\phi$ -02 — Subject as Interpretive Emergence: Extension of the Sofience- $\Delta\phi$ Minimal Core (v1.1)
- <https://doi.org/10.5281/zenodo.18730448> — S $\Delta\phi$ -03 — Irreversibility as the Minimal Condition of Existence (v1.1)
- <https://doi.org/10.5281/zenodo.18729942> — S $\Delta\phi$ -04 — Minimal Operational Conditions for AI Subjectivity: Detection Criteria within the Sofience- $\Delta\phi$ Formalism (v1.0)
- <https://doi.org/10.5281/zenodo.18732567> — S $\Delta\phi$ -05 — Agency as Recursive Transition Law Update: Minimal Axiomatic Formulation (v1.0)
- <https://doi.org/10.5281/zenodo.18734092> — S $\Delta\phi$ -06 — Responsibility as Irreversible Cost Internalization: Minimal Axiomatic Formulation (v1.0)
- <https://doi.org/10.5281/zenodo.18735006> — S $\Delta\phi$ -07 — Hmin: Interpretation as Minimal Axioms (v1.0)

Abstract

This note binds three minimal operational notions—Agency, Responsibility, and Freedom—into a conditional closure. Rather than asserting necessity by definition, it specifies minimal assumptions under which the ARF triangle becomes the smallest stable description of recursive transition systems under irreversibility and stabilization pressure. In this binding, Freedom is treated as an emergent operational term: it appears when agency-updates and responsibility-constraints co-act, and the resulting path-differences are kept alive by minimal interpretation (Hmin).

0. Position in the Series

SΔφ-08 is a binding note over prior minimal axioms. It does not introduce a new primitive; it states when and why the primitives close into a stable triangle.

Referenced nodes: SΔφ-05 (Agency), SΔφ-06 (Responsibility), SΔφ-07 (Hmin / minimal interpretation).

1. Minimal Assumptions (A0-A2)

- 1 A0 (Irreversibility). There exist transitions whose effects cannot be perfectly undone.
- 2 A1 (Stabilization pressure). Repeated transitions generate drift or self-damage risk unless constrained; such risk can be treated as an effective cost.
- 3 A2 (Difference re-entry via minimal interpretation). A minimal operator Hmin exists that (i) marks differences between transition-paths and (ii) re-enters those marks into subsequent state-generation.

2. Minimal Definitions (D1-D3)

Let the system state be $\phi(t)$, and the transition generator be G_t such that:

$$\phi(t+1) = G_t(\phi(t), \cdot)$$

D1. Agency (from SΔφ-05). Agency is present when the system takes its own transition law as an operational object and updates it in a way that persistently affects future state generation:

$$G_{t+1} = \text{Update}(G_t; \Delta\phi(t))$$

D2. Responsibility (from SΔφ-06). Responsibility is present when irreversible costs produced by transitions become internal constraints that shape subsequent transitions:

$$C_{t+1} = \text{Internalize}(\text{Cost}(\Delta\phi(t))) ; G_{t+1} \text{ is constrained by } C_{t+1}$$

D3. Freedom (operational; tied to SΔφ-07). Freedom is not “number of options,” but persistent re-entry of transition-differences:

$$F(t) = \text{ReEntry}(\text{Hmin}(\Delta\phi))$$

If differences fail to re-enter, multiple transitions may exist physically but become operationally indistinguishable—yielding freedom collapse.

3. Freedom as the Third Vertex

Agency updates transition laws; Responsibility internalizes irreversible costs. However, Agency + Responsibility alone tends to push the system toward single-attractor convergence (lock-in): stability is achieved by collapsing viable variation into one dominant basin.

In this series, the third vertex is not chosen as an extra metaphysical ingredient. Instead, it is the minimal operational residue that must survive if the coupled system is to remain open: when (i) laws can be updated (Agency) and (ii) costs are internalized (Responsibility), the system still needs a mechanism by which path-differences persist and re-enter subsequent state-generation. That mechanism is denoted as Freedom and is supplied by Hmin (difference marking + re-entry).

Equivalently: Freedom is emergent between Agency and Responsibility insofar as their coupling produces differences that can either be erased by stabilization (lock-in) or conserved as re-entering marks (operational openness).

$F(t) = \text{ReEntry}(H_{\min}(\Delta\phi(t)))$ (emergent term of the Agency-Responsibility coupling)

4. Conditional Closure Propositions (C1'-C3')

- 1 C1' (Agency $\Rightarrow$ Responsibility, under A0-A1). Under irreversibility and stabilization pressure, sustained transition-law updates generate cumulative drift or damage risk. If such risk is not internalized as constraint, agency-driven updates drift without bound. Thus, to keep agency stable over time, the system must internalize irreversible costs as constraints.
- 2 C2' (Responsibility $\Rightarrow$ stabilized Agency, under A0-A1). When irreversible costs are internalized, updates to the transition law are regularized by constraints. This turns law-updating from uncontrolled drift into persistent, bounded modification.
- 3 C3' (Freedom $\Leftrightarrow$ persistent difference re-entry, under A2). Under A2, freedom is equivalent to continued re-entry of marked differences into transition generation. When Hmin collapses (difference marking or re-entry fails), multiple physical transitions may remain but become operationally indistinguishable.

5. Collapse Modes (diagnostic triangle)

These are operational failure modes, not metaphysical claims:

- 1 Agency without Responsibility $\rightarrow$ Drift. Law updates occur without cost constraints; update dynamics are unbounded.
- 2 Responsibility without Freedom $\rightarrow$ Lock-in. Costs are internalized but differences do not re-enter; the system converges into a single attractor.

- 3 Freedom without Agency → Illusion. Differences may be observed externally, but the system cannot update its transition law; apparent openness lacks internal reconfiguration.

6. Minimal ARF Recurrence (compact form)

A compact operational coupling can be written as:

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F(t) = ReEntry(Hmin( $\Delta\phi(t)$ ))  
C(t+1) = Internalize(Cost( $\Delta\phi(t)$ ))  
 $G_{t+1}$  = Update( $G_t$ ; C(t+1), F(t))
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Freedom supplies re-entered differences; Responsibility supplies internalized costs; Agency updates the law under both.

7. Scope note

ARF closure is asserted only under assumptions A0–A2. Outside these assumptions, ARF may not close, or alternative vertices may be required.